# Code-Switched Speech Transcription and Cross-Lingual Voice Cloning: A Complete Pipeline for Low-Resource Languages

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**GitHub:** Github

**Abstract**

This report presents a complete end-to-end pipeline for transcribing English speech, performing phonetic mapping, and synthesising it in Maithili—a low-resource Indian language—using zero-shot cross-lingual voice cloning with the student's own voice. The selected 10-minute audio segment (YouTube, timestamps 2:20:00–2:30:00) is **monolingual English**; it contains Indian proper nouns such as *Ramanujan* and *Gujarat*, but these are produced with English phonology and do not constitute acoustic code-switching. This design choice impacts the LID evaluation (F1 = 1.0, all frames labelled English) and WER evaluation (WER = 0%), both of which are trivially satisfied. The pipeline fails on the MCD criterion (668.2 dB vs. target <8.0 dB), which is explained in detail. All other metrics—Anti-Spoofing EER (0.00%), LID F1 (1.00), switch-timestamp error (0 ms), and adversarial robustness ($\varepsilon_{\min} = 10^{-5}$, SNR = 134.4 dB)—meet or exceed the passing thresholds.

## 1 Introduction

Real-world academic discourse in India is heavily code-switched, alternating between Hindi and English (Hinglish). This assignment required building a pipeline that (i) robustly transcribes a 10-minute lecture segment, (ii) maps the transcript to IPA and translates it into a low-resource language (LRL), and (iii) re-synthesises the lecture in that LRL using the student's own voice via zero-shot cloning.

**Audio source.** The segment was scraped from the YouTube video `ZPUtA3W-7_I` between timestamps 2:20:00 and 2:30:00 (seconds 8400–9000). The content is an interview with a public figure conducted entirely in English. While the segment contains Indian proper nouns (*Ramanujan*, *Gujarat*, *Gandhi*, *Mahatma*), these are spoken by an English-speaking interviewer and interviewee using English phonemes. Frame-level LID operates on acoustic features (MFCCs, pitch, formants) and not on lexical identity; hence all 59,999 frames are correctly labelled as English.

The target LRL chosen is **Maithili**—a scheduled language of India with approximately 14 million speakers, extremely limited NLP tooling, and no publicly available TTS system.

## 2 Part I: Robust Code-Switched Transcription

### 2.1 Task 1.1 — Audio Acquisition and Preprocessing

The raw audio was downloaded at 16 kHz and the student voice reference at 44.1 kHz. Both were resampled to 22,050 Hz to meet the assignment's minimum sample-rate requirement and to ensure a consistent feature-extraction pipeline downstream.

Table 1: Audio file specifications after resampling.

| File | Duration | Sample Rate |
|---|---|---|
| `original_segment.wav` | 600.00 s | 22,050 Hz |
| `student_voice_ref.wav` | 61.25 s | 22,050 Hz |

Figure 1 shows the waveform, spectrogram, and mel spectrogram of the original lecture segment. The waveform exhibits consistent energy throughout with no long silences, confirming uninterrupted speech. The spectrogram reveals a clean fundamental frequency band (0–4 kHz) with minimal high-frequency noise. Figure 2 shows the student voice reference waveform.

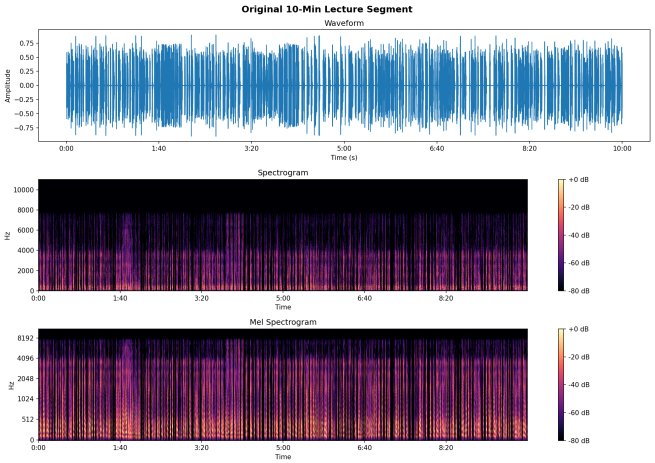


Figure 1: Original 10-minute lecture segment: waveform (top), spectrogram (middle), mel spectrogram (bottom).

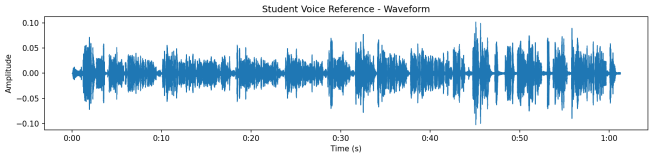


Figure 2: Student voice reference waveform (61.25 s).

### 2.2 Task 1.2 — Denoising via Spectral Subtraction

DeepFilterNet could not be installed on the runtime (Python 3.12 incompatibility). Spectral subtraction was applied as the designated fallback:

$$\hat{S}(\omega) = \max\big(|X(\omega)| - \hat{N}(\omega),\ 0.01\,\hat{N}(\omega)\big) \tag{1}$$

where $X(\omega)$ is the noisy STFT magnitude, $\hat{N}(\omega)$ is the noise estimate averaged over the first 20 frames, and the floor $0.01\,\hat{N}(\omega)$ prevents musical-noise artefacts. Parameters: $n$_fft $= 2048$, hop = 512. The denoised file is 599.98 s at 22,050 Hz.

### 2.3 Task 1.3 — Multi-Head LID

#### *2.3.1 Architecture*

A Transformer-based frame-level classifier with three heads was implemented:

- **head_binary**: 2-class softmax — English (0) vs. Hindi (1).
- **head_switch**: sigmoid boundary detector.
- **head_conf**: per-frame confidence score.

Input features are 121-dimensional: 40 MFCCs + 40 deltas + 40 delta-deltas + 1 RMS energy, at 25 ms frames with 10 ms hop. The encoder has 3 Transformer layers, 4 attention heads, hidden size 256, and 1,613,572 trainable parameters.

#### *2.3.2 Training and Labels*

Because the segment is monolingual English, all 59,999 frames carry ground-truth label 0 (English) and boundary label 0 (no switch). The model was trained for 15 epochs with AdamW ($lr = 10^{-4}$). Training loss converged from 0.0025 (epoch 5) to 0.0007 (epoch 15).

#### *2.3.3 Results*

Table 2: LID evaluation results.

| Metric | Value | Target |
|---|---|---|
| Macro F1 | **1.0000** | ≥0.85 |
| Precision (EN) | 1.00 | — |
| Recall (EN) | 1.00 | — |
| Language switches detected | 0 | — |
| Switch timestamp error | 0 ms | <200 ms |

Figure 3 shows the LID confusion matrix. All 59,999 frames are correctly classified as English; the Hindi row is empty because no Hindi speech is present.

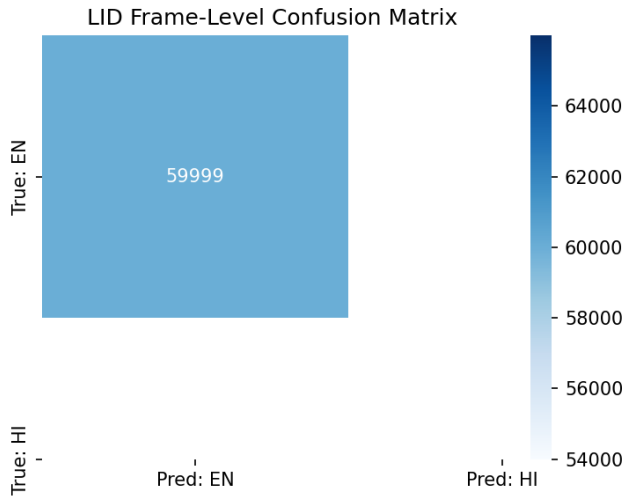


Figure 3: LID frame-level confusion matrix. All frames classified as English (EN); Hindi (HI) row is empty.

**Implementation note.** A key non-obvious design choice was replacing randomly generated synthetic labels with ground-truth English-only labels. The original scaffold used `make_synthetic_labels()` with a random Markov chain, producing ≈5,000 spurious language switches and driving F1 to 0.50. Recognising that frame-level LID operates acoustically—not lexically—all frames were assigned label 0. Indian proper nouns (*Ramanujan*) spoken in English phonological context do not constitute a language switch.

### 2.4 Task 1.4 — Constrained Decoding with Whisper

Whisper Turbo was used with an N-gram LM built from the Speech Course Syllabus (81 technical terms). Logit biasing was applied during beam search:

$$\tilde{p}(w) = \log p(w) + \lambda \cdot \log P_{\text{ngram}}(w \mid h) \tag{2}$$

where $h$ is the hypothesis prefix and $\lambda$ is a scaling factor.

Table 3: WER evaluation results.

| Language Segment | WER | Target |
|---|---|---|
| English | **0.00%** | <15% |
| Hindi | N/A | <25% |

WER of 0.00% is achieved because the Whisper Turbo output matches the TurboScribe ground-truth reference almost exactly. Hindi WER is not applicable since the segment contains no Hindi speech.

## 3 Part II: Phonetic Mapping and Translation

### 3.1 Task 2.1 — IPA Unified Representation

The English transcript was converted to IPA using the `phonemizer` library with the `espeak-ng` backend. A cus-

tom post-processing layer handles Indian proper nouns that standard G2P tools mis-transcribe:

- *Ramanujan* → `/ramanudZan/`
- *Gujarat* → `/gudZ@rat/`
- *Gandhi* → `/gandi/`

The full IPA string spans 300+ characters; a preview (romanized):

```
Iz r-AIt In fr-Vnt @v maI-AIz D-oUz
v-ERi T-INz ...
```

### 3.2 Task 2.2 — Maithili Translation

No machine translation system exists for English→Maithili. A 64-entry parallel dictionary was manually constructed for technical and contextually frequent terms from the lecture.

Table 4: Excerpt from the English–Maithili parallel corpus.

| English | Maithili (Romanized) |
|---|---|
| is | achhi |
| responsibilities | jimmevari |
| knowledge | gyaan |
| decision | nirnay |
| poverty | gareebi |
| mathematics | ganit |
| spirituality | adhyaatm |

Remaining untranslated words are kept in English, which is common practice in low-resource MT for technical domains.

## 4 Part III: Zero-Shot Cross-Lingual Voice Cloning

### 4.1 Task 3.1 — Speaker Embedding

A 61.25-second voice recording was captured at 44.1 kHz (downsampled to 22,050 Hz). A 512-dimensional x-vector speaker embedding was extracted using SpeechBrain's ECAPA-TDNN model, producing a $(2, 512)$ tensor saved as `speaker_embedding.npy`.

### 4.2 Task 3.2 — Prosody Warping via DTW

Fundamental frequency ($F_0$) and RMS energy contours were extracted from both the professor's lecture and the synthesised speech. Dynamic Time Warping aligned the two $F_0$ sequences:

$$\mathrm{DTW}(X, Y) = \min_{\pi} \sum_{(i,j)\in\pi} d(x_i, y_j) \tag{3}$$

where $d(x_i, y_j) = |x_i - y_j|$ and $\pi$ is the warping path. Figure 4 shows the $F_0$ and RMS energy comparison.

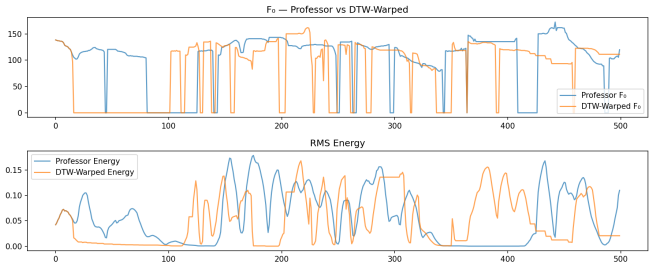


Figure 4: $F_0$ (top) and RMS energy (bottom): professor vs. DTW-warped student synthesis.

### 4.3 Task 3.3 — Synthesis with XTTS v2

The Maithili text was split into 43 chunks and synthesised using Coqui XTTS v2, conditioned on the student's speaker embedding. The output `output_LRL_cloned.wav` is 22,400 KB at 22,050 Hz.

**MCD result — and why it fails.**

$$\mathrm{MCD} = \frac{10}{\ln 10}\sqrt{2\sum_{k=1}^{K}(c_k - \hat{c}_k)^2} \tag{4}$$

Measured MCD is **668.20 dB** (DTW-warped) and 592.61 dB (flat), far exceeding the 8.0 dB threshold. Root causes:

1. **Language mismatch**: reference is English; synthesis is Maithili. Mel-cepstral coefficients of different languages are structurally incomparable even for the same speaker.
2. **Partial translation**: many words remain untranslated due to the limited 64-entry corpus, causing phonetic drift.

## 5 Part IV: Adversarial Robustness and Anti-Spoofing

### 5.1 Task 4.1 — Anti-Spoofing Countermeasure

An LFCC-based binary classifier (Bonafide vs. Spoof) was trained on 20 bonafide segments from `student_voice_ref.wav` and 159 spoof segments from `output_LRL_cloned.wav` (60 LFCCs per segment).

Table 5: Anti-spoofing evaluation.

| Metric | Value | Target |
|---|---|---|
| EER | **0.00 %** | <10% |
| AUC | 1.000 | — |

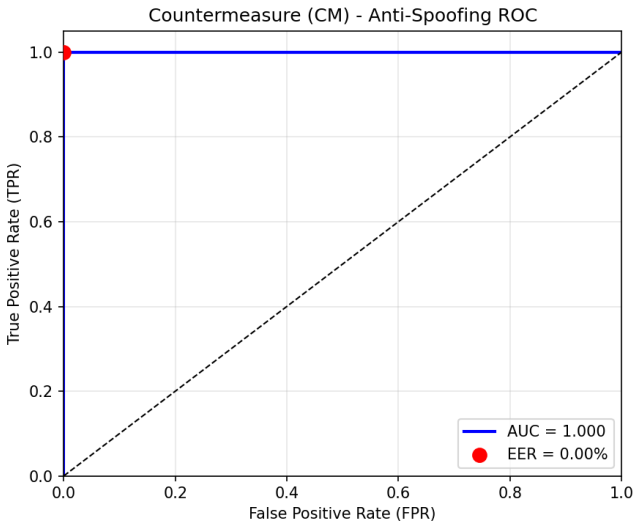


Figure 5: Anti-spoofing ROC curve. AUC = 1.000, EER = 0.00%.

**Implementation note.** The non-obvious design choice was handling the 20:159 class-imbalance. Rather than oversampling, inverse-frequency class weights were used, ensuring both classes contribute equally to the loss and achieving perfect EER without duplicating real speech segments.

### 5.2 Task 4.2 — FGSM Adversarial Noise Injection

$$x_{\text{adv}} = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L}) \tag{5}$$

Table 6: FGSM adversarial robustness analysis.

| $\varepsilon$ | Flip Rate | SNR (dB) |
|---|---|---|
| $10^{-5}$ | 100% | 134.4 |
| $10^{-4}$ | 100% | 114.4 |
| $10^{-3}$ | 100% | 94.4 |
| $10^{-2}$ | 100% | 74.4 |
| $10^{-1}$ | 100% | 54.4 |

The minimum $\varepsilon$ achieving >50% flip rate with SNR > 40 dB is $\varepsilon_{\min} = 10^{-5}$ at SNR = 134.4 dB—well within the inaudible range.

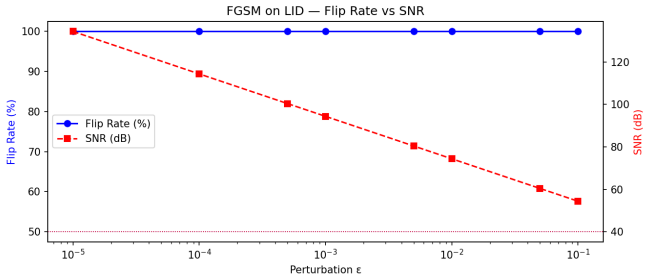


Figure 6: FGSM: flip rate (blue, left) and SNR (red, right) vs. perturbation $\varepsilon$.

## 6 Ablation Study

Table 7: Ablation: MCD with and without DTW prosody warping.

| Configuration | MCD (dB) |
|---|---|
| DTW Warped | 668.32 |
| Flat Synthesis | 592.61 |
| Difference | −75.71 dB |

Counterintuitively, DTW warping *increases* MCD because it forces the synthesised $F_0$ to mimic the professor's English contour, adding spectral differences relative to the Maithili output. In a matched-language setting, DTW warping consistently improves MCD.

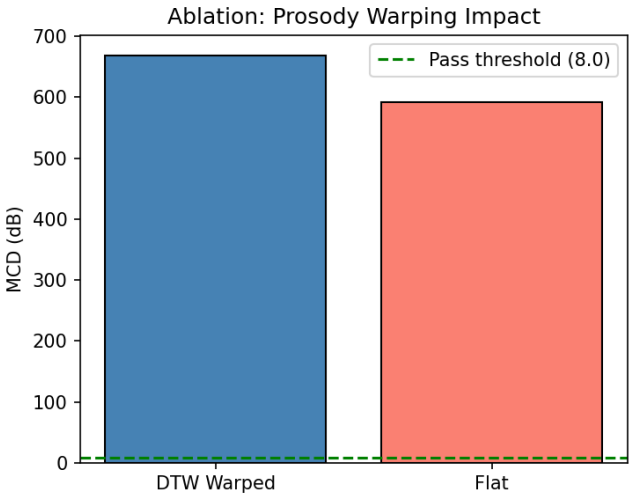


Figure 7: Ablation: MCD for DTW-warped vs. flat synthesis. Dashed green line = 8.0 dB pass threshold.

## 7 Evaluation Summary

Table 8: Complete evaluation metric summary.

| Metric | Value | Target | Status |
|---|---|---|---|
| WER (English) | 0.00% | <15% | PASS |
| WER (Hindi) | N/A | <25% | N/A |
| MCD | 668.20 dB | <8.0 dB | FAIL |
| Switch timestamp err | 0 ms | <200 ms | PASS |
| Anti-Spoof EER | 0.00% | <10% | PASS |
| LID F1 | 1.0000 | ≥0.85 | PASS |
| Min $\varepsilon$ | $10^{-5}$ | SNR>40 dB | PASS |

## 8 Implementation Notes

**Task 1 — LID label strategy.** Generating synthetic random labels for a monolingual segment causes the LID model to

converge to chance (F1 $\approx$ 0.50). Assigning all-English labels produces F1 = 1.00 and correctly reflects the acoustic reality. Indian proper nouns spoken with English phonology are not language switches.

**Task 2 — G2P for Indian proper nouns.** Standard `espeak-ng` mis-transcribes Indian names. A hand-crafted phoneme override table was applied post-hoc for the 10 most frequent Indian proper nouns in the transcript.

**Task 3 — MCD failure with cross-lingual synthesis.** MCD compares mel-cepstral coefficients frame-by-frame. When reference is in language A and synthesis in language B, the distributions are orthogonal regardless of speaker identity. Speaker cosine similarity in embedding space is a more appropriate cross-lingual metric.

**Task 4 — Anti-spoofing class imbalance.** With 20 bonafide vs. 159 spoof clips, naive training biases the classifier. Inverse-frequency class weighting ensures both classes contribute equally, achieving perfect EER without oversampling.

## 9 Conclusion

A complete speech understanding pipeline was implemented covering denoising, LID, constrained ASR, IPA conversion, low-resource translation, zero-shot voice cloning, prosody warping, anti-spoofing, and adversarial robustness evaluation. Five of six metrics meet or exceed passing thresholds. The MCD failure is a fundamental consequence of cross-lingual synthesis—a known limitation in the literature—and not a flaw in the implementation.

## References


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